

Code of practice for learning analytics

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Learning analytics refers to the measurement, collection, analysis and reporting of data about learning, learning environments and student engagement. The original definition came from the 1st International Conference on Learning Analytics and Knowledge. **solaresearch.org/about/what-is-learning-analytics/**



Introduction

Students using digital resources and systems generate data that can be analysed to reveal patterns predicting success, difficulty or failure, which enable teachers - and students - to make timely interventions. These metrics can also support a more accurate, data-informed approach to curriculum design. It is distinct from assessment and should be used for formative rather than summative purposes.

Learning analytics should be used for the benefit of students. This might be to assist them individually or through using aggregated and anonymised data to help other students.

The effective use of learning analytics will initially involve the deployment of new systems along with changes to institutional policies and processes. New data will be collected about individuals and their learning activities. The data will be analysed and interventions may take place as a result. This presents opportunities for positive, evidence-led interventions.

Complete transparency and clear institutional policies are essential

Complete transparency and clear institutional policies are therefore essential regarding the purposes of learning analytics, the data collected, the processes involved, and how they will be used to enhance the educational experience.

This code of practice aims to set out the responsibilities of educational institutions to ensure that learning analytics is carried out responsibly, appropriately and effectively, addressing the key legal, ethical and logistical issues which are likely to arise.

Educational institutions in the UK already have information management practices and procedures in place and have extensive experience of handling sensitive and personal data in accordance with the General Data Protection Regulation (GDPR). The UK data protection regime is regulated by the Data Protection Act 2018 and the GDPR has been written into UK law and tailored to become the 'UK GDPR.'

By transferring and adapting this expertise to regulate the processing of data for learning analytics, institutions should establish the practices and procedures necessary to process the data of individuals lawfully and fairly.

Responsibility

Institutions must decide who has overall responsibility for the legal, ethical and effective use of learning analytics. Examples of roles who would be ultimately responsible for learning analytics are University Registrar, Dean of Students, Deputy VC or Pro VC.

They should allocate specific responsibility within the institution for:

- The collection of data to be used for learning analytics
- The anonymisation of the data where appropriate
- The analytics processes to be performed on the data, and their purposes
- The interventions to be carried out
- The retention and stewardship of data used for and generated by learning analytics

Student representatives and key staff groups at institutions should be consulted about the objectives, design, development, roll-out and monitoring of learning analytics.

Appropriate University staff may suggest actions for consideration by students to help improve their progress. Students will have responsibility for deciding the best application of any suggestions to their learning approach.

As a minimum, anyone accessing learning analytics data should complete Information Security training.

Transparency, legal basis and consent

Institutions will define the objectives for the use of learning analytics, what data is necessary to achieve these objectives, and what is out of scope. The data sources, the purposes of the analytics, the metrics used, who has access to the analytics, the boundaries around usage and how to interpret the data must be explained clearly to staff and students.

Institutions should also clearly describe the processes involved in producing the analytics to students and staff or make the algorithms transparent to them.

Consider these policies and guidance to support transparency:

- Student Charter
- Student guide to learning analytics
- Learning analytics policy
- Engagement policy
- Intervention policy

The primary legal basis for the use of a learning analytics service is the institution's legitimate interest in providing the best possible educational experience for its students. Where students provide their own additional data, for example through a learning analytics student application, the most likely basis is the student's free, informed consent.

Students will normally be asked for their consent for personal interventions to be taken based on the learning analytics. This will usually be done at the point where a specific intervention is proposed. However, there may be legal, safeguarding or other circumstances where students are not permitted to opt out of such interventions. If so, these must be clearly stated and justified.

Student consent is also required for the collection and use of “special category data”, such as ethnicity, as defined by GDPR.

New learning analytics projects may not be covered by the institution’s existing arrangements. Collection and use of data for these may require further measures, such as data protection impact assessments and obtaining additional consent.

Options for granting consent must be clear and meaningful, explaining the expected consequences of granting or withholding consent. Students should be able easily to amend their decisions subsequently.



Privacy

Access to student data and analytics should be restricted to those identified by the institution as having a legitimate need to view them.

Where data is to be used anonymously particular care must be taken by institutions to avoid:

- Identification of individuals from metadata
- Re-identification of individuals by aggregating multiple data sources

The use of "special category data" for the purposes of learning analytics requires additional safeguards. Circumstances where data and analytics could be shared externally - eg requests from educational authorities, security agencies or employers - will be made explicit to staff and students, and may require additional consent.

Institutions should ensure that student data is protected when third parties are contracted to store data or carry out learning analytics on it.

Institutions may have a legal obligation to intervene, and hence override some privacy restrictions, where data or analytics reveal that a student is at risk. Such circumstances should be clearly specified.

Validity

It is vital that institutions monitor the quality, robustness and validity of their data and analytics processes to develop and maintain confidence in learning analytics and ensure it is used to the benefit of students. Institutions should ensure that:

- Inaccuracies in the data are understood and minimised
- The implications of incomplete datasets are understood
- The optimum range of data sources is selected
- Spurious correlations are avoided

All algorithms and metrics used for predictive analytics or interventions should be understood, validated, reviewed and improved as appropriate by qualified staff.

Data and analytics may be valid but should also be useful and appropriate; learning analytics should be seen in its wider context and combined with other data and approaches as appropriate.

Understanding gained from interactions and relationships between staff and students within a learning context must be a significant contributor to any intervention.

Access

Students should be able to access all learning analytics performed on their data in meaningful, accessible formats, and to obtain copies of this data in a portable digital format. This is normally in the form of a Subject Access Request. Students have a legal right under the UK GDPR to be able to correct inaccurate personal data held about themselves.

If an institution considers that the analytics may have a harmful impact on the student's academic progress or wellbeing it may withhold the analytics from the student by default, subject to clearly defined and explained policies. However, the student has a legal right and must be shown the data about them if they ask to see it; in such circumstances we suggest this might be best affected in tandem with a conversation with a personal tutor, rather than via a wholly automated response.

Minimising adverse impacts

Institutions must recognise that analytics can never give a complete picture of an individual's learning and may sometimes ignore personal circumstances.

Institutions should take steps to ensure that trends, norms, categorisation or any labelling of students do not bias staff, student or institutional perceptions and behaviours towards them or reinforce discriminatory attitudes.

Analytics systems and interventions should be carefully designed and regularly reviewed to ensure that:

- Students maintain appropriate levels of autonomy in decision making relating to their learning, using learning analytics where appropriate to help inform their decisions
- Opportunities for "gaming the system" or any benefit to the student from doing so are minimised
- Knowledge that their activity is being monitored does not lead to non-participation by students or other negative impacts on their academic progress or wellbeing
- Adverse impacts as a result of giving students and staff information about the students' performance or likelihood of success are minimised
- Staff have a working understanding of legal, ethical and unethical practice

Stewardship of data

Data for learning analytics must comply with existing institutional data policies and the GDPR. In particular the data must be:

- Kept to the minimum necessary to deliver the purposes of the analytics reliably
- Processed in the UK, European Economic Area or, if elsewhere, only in accordance with the GDPR
- Retained only for appropriate and clearly defined periods (this should be defined within a formal retention schedule)

On request by students any personal data used for or generated by learning analytics should be destroyed or anonymised, except for certain, clearly specified data fields required for educational or statutory purposes such as grades.

Moving towards wellbeing analytics

Whereas learning analytics uses data to inform decisions on how to support students' learning, data may also be used to inform decisions on how to support wellbeing and mental health. See our companion **code of practice for wellbeing and mental health analytics.**

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